LUMEXA
Space-themed coding education for K–12

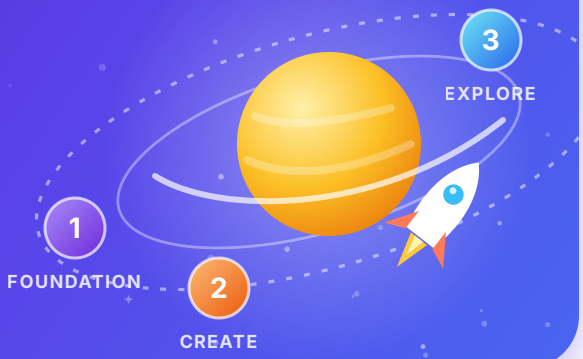
AGES 6–18 | BEST CHOICE FOR BEGINNERS | SUITABLE FOR ALL AGES

FLAGSHIP CURRICULUM

Lumexa Odyssey

72 Lessons. One Big Launchpad.

The first path every new student takes. Odyssey is a guided journey across nine real courses — games, websites, apps, data, and AI — giving your child broad, hands-on exposure before they choose a specialty. No prior coding experience needed.



72 LESSONS · 9 REAL COURSES
Scratch, Python, the web, data and AI

13 TESTS ALONG THE WAY
9 course tests + 3 stage tests + 1 final test

9	72	85	6–18
Courses	Lessons	Sessions in total	Age range

WHO IT IS FOR
Suitable for all ages — the best choice for beginners

BEFORE YOU START
No coding experience needed

THE ODYSSEY JOURNEY

STAGE 1 · FOUNDATION
1. Scratch Adventures
2. Python for Young Builders
3. Python Arcade Games

STAGE 2 · CREATE & APPLY
4. HTML and CSS Mastery
5. Build Your Professional Identity
6. JavaScript and Interactivity

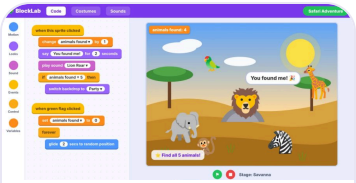
STAGE 3 · EXPLORE & ADVANCE
7. Python for Data
8. Python and AI Foundations
9. Roblox World Builder

Breadth before depth
Nine real courses across games, websites, data and AI — so students choose a specialty from experience, not guesswork.

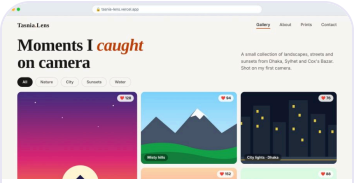
Real projects at every step
Students build and show finished work in Scratch, Python, HTML/CSS and JavaScript, plus a live portfolio and a 3D world.

Tested, so progress is real
Every course ends with a test, and a final test closes the program, so progress is proven, not assumed.

SAMPLE HANDS-ON PROJECTS STUDENTS BUILD



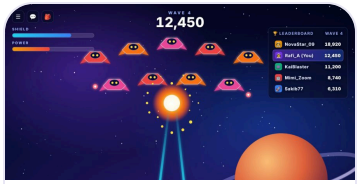
01 Safari Adventure
Scratch-style blocks



04 Photography Portfolio
HTML · CSS



07 Sports Stats Analyser
Python · Pandas



09 Galactic Shooter
Roblox · Lua

STAGE 1 OF 3 · LESSONS 1–24

1 Foundation

Every great builder starts somewhere. Stage 1 turns total beginners into confident, independent coders through games and playful logic.

3 COURSES

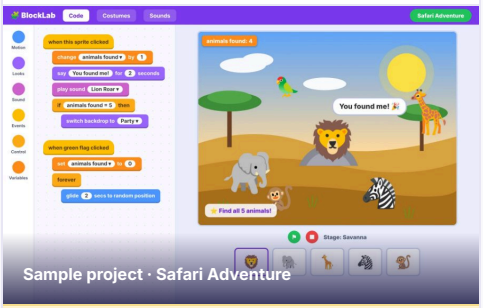
3 COURSE TESTS + 1 STAGE TEST

COURSE 01 · LESSONS 1–8 — Scratch Adventures

Students build their first interactive stories and games in Scratch's drag-and-drop editor — learning sequencing, loops, and events through play.

Scratch · Sequencing · Loops & events

1. What Is Coding, Anyway?
2. Meet Scratch and Your First Sprite
3. Motion, Looks, and Sound — Your Sprite's 3 Superpowers
4. Events and Broadcasting — Sprites That React!
5. Loops — Do It Again (and Again)!
6. If This, Then That — Sprites That Decide!
7. Build Your Own Interactive Story
8. Final Project — Build Your Own Game!



Sample project · Safari Adventure

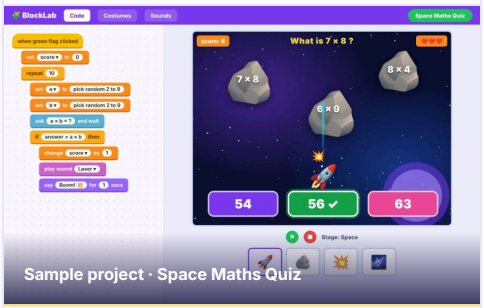
Scratch Adventures — Course Test

COURSE 02 · LESSONS 9–16 — Python for Young Builders

A gentle, guided introduction to real text-based programming in Python — variables, conditionals, and loops, applied to small practical tools.

Python 3 · Variables · Conditionals · Loops

1. From Blocks to Words — Hello, Python!
2. Variables and User Input
3. Printing and String Formatting
4. If/Else Decisions
5. While Loops and For Loops
6. Lists and Simple Data
7. Building a Mini Quiz App
8. Final Project — Our Own Mini Game!



Sample project · Space Maths Quiz

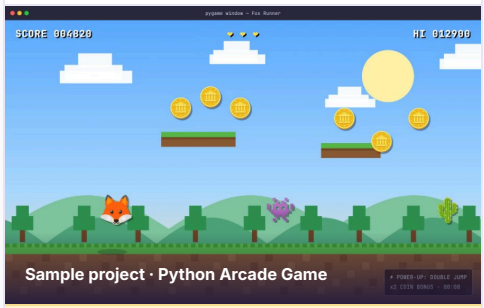
Python for Young Builders — Course Test

COURSE 03 · LESSONS 17–24 — Python Arcade Games

Students apply their new Python skills to build real, playable arcade-style games — turning foundational logic into something they can show off.

Python · Pygame · Game development

1. Python Basics for Game Development
2. Pygame Setup and the Game Loop
3. Sprites, Images, and Movement
4. Collision Detection
5. Enemies, AI Behavior, and Scoring
6. Sound Effects and Music
7. Game States — Menu, Play, Game Over
8. Packaging and Sharing Your Game



Sample project · Python Arcade Game

Python Arcade Games — Course Test

STAGE TEST · AFTER LESSON 24

Stage 1 Test

At the end of each stage, a stage test looks across the three courses just finished, so nothing is left behind.

30 multiple-choice · 10 true / false · 40 marks

STAGE 2 OF 3 · LESSONS 25–48

2 Create & Apply

With coding fundamentals in place, students start building real things for the real internet — and start presenting themselves professionally along the way.

3 COURSES

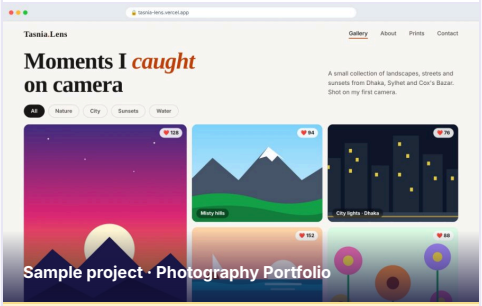
3 COURSE TESTS + 1 STAGE TEST

COURSE 04 · LESSONS 25–32 — HTML and CSS Mastery

Students learn to structure and style real webpages from scratch — the foundational language of every website on the internet.

HTML5 · CSS3 · Web design basics

1. How the web works (browsers, servers, HTML)
2. HTML structure and semantic elements
3. CSS basics: colors, fonts, spacing
4. Flexbox for modern layouts
5. CSS Grid for complex designs
6. Responsive design and mobile-first
7. Animations and hover effects
8. Deploying to Vercel and sharing



Sample project · Photography Portfolio

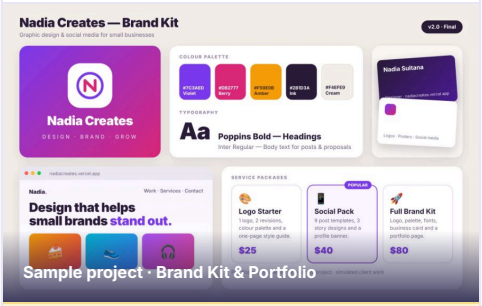
HTML and CSS Mastery — Course Test

COURSE 05 · LESSONS 33–40 — Build Your Professional Identity

Students build a live personal portfolio site and learn to present their growing skills honestly and professionally online.

Portfolio site · Professional identity

1. What Professional Identity Means in the Digital Age
2. Define Your Strengths and Direction
3. Build Your Portfolio Website
4. Tell Your Story Professionally
5. Create Your Personal Brand
6. Turn Projects Into Proof
7. Launch Your Professional Website
8. Build Social Proof



Sample project · Brand Kit & Portfolio

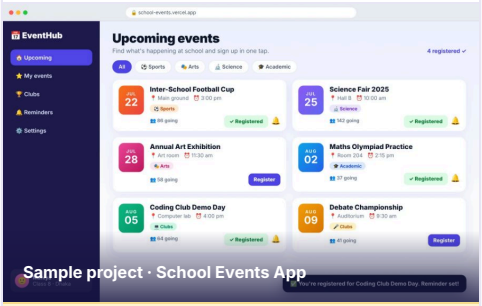
Build Your Professional Identity — Course Test

COURSE 06 · LESSONS 41–48 — JavaScript and Interactivity

Students bring their websites to life — adding buttons, animations, and interactive logic using real JavaScript.

JavaScript · Interactive web

1. JavaScript fundamentals (variables, functions, loops)
2. DOM manipulation: selecting and changing elements
3. Event listeners: responding to user actions
4. Fetch API: pulling live data from the web
5. Async/await and working with APIs
6. Local Storage for saving user data
7. Building a complete interactive app
8. Debugging and polishing



Sample project · School Events App

JavaScript and Interactivity — Course Test

STAGE TEST · AFTER LESSON 48

Stage 2 Test

At the end of each stage, a stage test looks across the three courses just finished, so nothing is left behind.

30 multiple-choice · 10 true / false · 40 marks

STAGE 3 OF 3 · LESSONS 49–72

3 Explore & Advance

A guided tour of the paths ahead — data, artificial intelligence, and 3D world building — so students choose their specialty with real experience, not guesswork.

3 COURSES

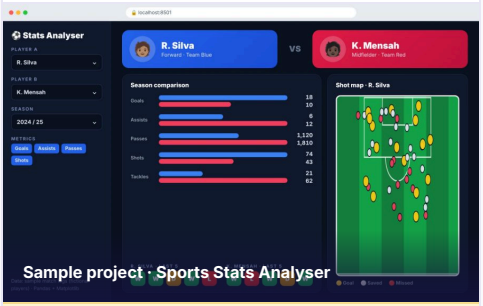
3 COURSE TESTS + 1 STAGE TEST

COURSE 07 · LESSONS 49–56 — Python for Data (07)

Students learn to organize, analyze, and visualize real data using Python — a first taste of the data scientist's toolkit.

Python · Data basics

1. Why Data Science Matters (Real Examples)
2. Python Refresher and Jupyter Setup
3. Loading Data from CSV and Excel
4. Pandas — Selecting, Filtering, Sorting
5. Cleaning Data — Nulls, Duplicates, Types
6. Aggregations — Group By, Pivot Tables
7. Statistical Summaries and Correlation
8. Full Analysis Project with a Real Dataset



Sample project · Sports Stats Analyser

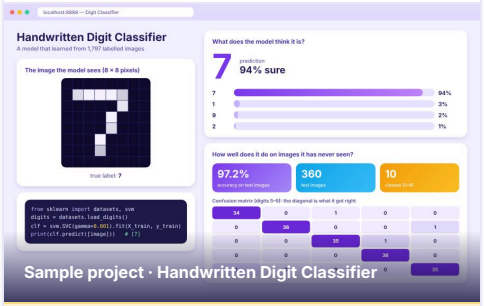
Python for Data — Course Test

COURSE 08 · LESSONS 57–64 — Python and AI Foundations (08)

An approachable introduction to how artificial intelligence actually works — students build and experiment with simple, real AI models.

Python · AI foundations

1. Python Setup and First Program
2. Variables, Data Types, and Logic
3. Loops, Functions, and Modules
4. Working with Data and Lists
5. Introduction to Machine Learning Concepts
6. Building a First Prediction Model
7. Training, Testing, and Evaluating Models
8. Presenting Results to Non-Technical Audiences



Sample project · Handwritten Digit Classifier

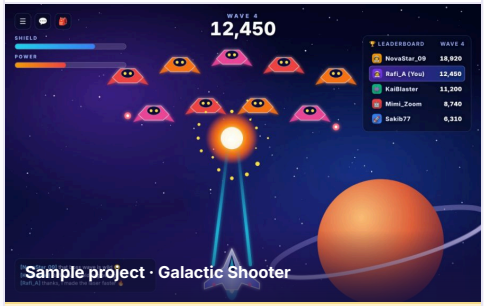
Python and AI Foundations — Course Test

COURSE 09 · LESSONS 65–72 — Roblox World Builder (09)

Students design and script an interactive 3D world in Roblox Studio — combining creativity, logic, and game design in one capstone-style course.

Roblox Studio · 3D & game design

1. Roblox Studio Setup and First Build
2. Scripting Basics in Lua
3. Player Mechanics and Movement
4. Game Logic and Scoring Systems
5. Multiplayer Networking Basics
6. World Design and Aesthetics
7. Testing, Debugging, and Polishing
8. Publishing and Sharing with Friends



Sample project · Galactic Shooter

Roblox World Builder — Course Test

STAGE TEST · AFTER LESSON 72

Stage 3 Test

At the end of each stage, a stage test looks across the three courses just finished, so nothing is left behind.

30 multiple-choice · 10 true / false · 40 marks

HOW STUDENTS ARE TESTED · 13 TESTS

Level-Up Checkpoints

Odyssey is tested the whole way: a course test after every 8 lessons, a stage test after every three courses, and one final test over the entire journey — 13 checkpoints that show a student is ready to choose a specialty.



THE CHECKPOINT PATH — EVERY COURSE ENDS WITH A TEST, THEN ONE FINAL

Stage 1 · 24 lessons
3 course tests
STAGE 1 TEST

Stage 2 · 24 lessons
3 course tests
STAGE 2 TEST

Stage 3 · 24 lessons
3 course tests
STAGE 3 TEST

FINAL

COURSE TEST · every 8 lessons

Show what you built and learned

After every 8 lessons a course test checks the ideas and skills of that course, so each next course starts on solid ground.

30	multiple-choice questions — choose one of four answers
10	true / false questions
40	marks in total — 1 mark for every question

STAGE TEST · every stage

Prove a whole stage

At the end of each stage, a stage test looks across the three courses just finished, so nothing is left behind.

30	multiple-choice questions across the stage
10	true / false questions
40	marks in total — 1 mark for every question

FINAL TEST · the finish line

One final test on the whole curriculum

The finish line of the Odyssey: questions across all nine courses and all three stages, plus a small practical project.

25	multiple-choice questions
5	true / false questions
20	marks for one small practical project (about 15 minutes) that your teacher marks

A TASTE OF A TEST

SAMPLE · MULTIPLE-CHOICE

Which language gives a web page its structure?

- A HTML ✓ CORRECT
- B Lua
- C Pandas
- D Scratch

SAMPLE · TRUE / FALSE

A loop lets a program repeat steps without writing them out again.

True ✓ | False

Examples in the style of a real test — written for this document, not taken from one.

AFTER EVERY TEST

Saved as you go
Answers save automatically while a student works, so nothing is lost if the connection drops.

Checked by the teacher
The teacher reviews the report and shares the result with the student.

Fair for everyone
Questions and answer options appear in a different order for every student.

The Complete Journey

Across 9 courses, 72 lessons and 13 tests, students move from total beginner to confident, well-rounded builder — with hands-on experience across games, the web, data, and AI, all before choosing where to specialize.

9 COURSES · 72 LESSONS

STAGE 1 · FOUNDATION

- 01 **Scratch Adventures** — Scratch · Stories & games
- 02 **Python for Young Builders** — Python · Quiz app & mini game
- 03 **Python Arcade Games** — Python · Arcade games

3 course tests + stage test after lesson 24

STAGE 2 · CREATE & APPLY

- 04 **HTML and CSS Mastery** — HTML & CSS · Real webpages
- 05 **Build Your Professional Identity** — Portfolio · Professional identity
- 06 **JavaScript and Interactivity** — JavaScript · Interactive web

3 course tests + stage test after lesson 48

STAGE 3 · EXPLORE & ADVANCE

- 07 **Python for Data** — Python · Real data
- 08 **Python and AI Foundations** — Python · AI foundations
- 09 **Roblox World Builder** — Roblox Studio · 3D world & game design

3 course tests + stage test after lesson 72

HOW STUDENTS ARE TESTED · 13 TESTS

- COURSE TEST 1 — Scratch Adventures
- COURSE TEST 2 — Python for Young Builders
- COURSE TEST 3 — Python Arcade Games
- COURSE TEST 4 — HTML and CSS Mastery
- COURSE TEST 5 — Build Your Professional Identity
- COURSE TEST 6 — JavaScript and Interactivity
- COURSE TEST 7 — Python for Data
- COURSE TEST 8 — Python and AI Foundations
- COURSE TEST 9 — Roblox World Builder

- STAGE TEST 1 — Stage 1 · Foundation
- STAGE TEST 2 — Stage 2 · Create & Apply
- STAGE TEST 3 — Stage 3 · Explore & Advance

FINAL TEST — **One final test on the whole curriculum**
The finish line of the Odyssey: questions across all nine courses and all three stages, plus a small practical project.

BY THE END, STUDENTS CAN

- Write and debug real code across multiple languages — Scratch, Python, HTML/CSS, and JavaScript.
- Build and show off real, finished projects: games, a live website, a personal portfolio, and a 3D world.
- Confidently choose a specialized 24-lesson pathway based on real hands-on experience, not guesswork.

WHAT COMES NEXT — CHOOSE A SPECIALIZED PATHWAY

Game Creator · AI Builder · Web Developer · Data Scientist · Digital Independence

The best first step into coding, for any beginner.
All lessons and projects are taught live by an expert Lumexa instructor.

Lumexa Odyssey · Ages 6–18